

TRANSORG AGENTIQ DATATHON · TRACK 3 : AGRITECH

Mandi-to-Market Supply Chain Optimizer

Turning fragmented mandi arrivals, prices, MSP, weather and transport data into one intelligence layer for India's agricultural supply chains.

Insight Squad

The Problem

Agricultural supply chains generate rich but disconnected data: crop arrivals, wholesale prices, MSP, weather conditions and transport logistics.

Analysed separately, these datasets hide the real story.

It becomes difficult to see which mandis are actually under stress — and why.

Our project brings all five together and answers one question:

Which mandis need attention — and why?

FIVE DISCONNECTED DATASETS

1

Mandi Master

2

Mandi Arrivals

3

Price & MSP

4

Weather Sensors

5

Transport Logistics

What We Built

Data preprocessing, supply-chain analysis, mandi risk scoring, an AI agent and a live dashboard — combined into one workflow.

01

Data Cleaning & Integration

Merged five raw datasets into one consistent, analysis-ready supply-chain view.

02

Supply-Chain Risk Analysis

Computed price, arrival, weather and transport indicators for every mandi.

03

Mandi Priority Scoring

A weighted score ranks every mandi Critical → High → Medium → Low.

04

AI Data Analysis Agent

Natural-language Q&A over the full integrated dataset — no code required.

05

Interactive Dashboard

A live, web-based view of arrivals, prices, weather, transport and risk.

Data We Used

Five datasets provided for the challenge, cleaned and joined into a single supply-chain view.

1

Mandi Master

Mandi, district, state and mandi-type information

2

Mandi Arrivals

Crop arrivals and quantities

3

Price & MSP

Wholesale prices vs. Minimum Support Price

4

Weather Sensors

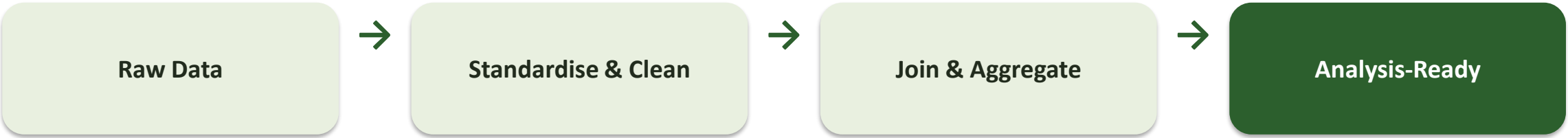
Temperature, rainfall and humidity

5

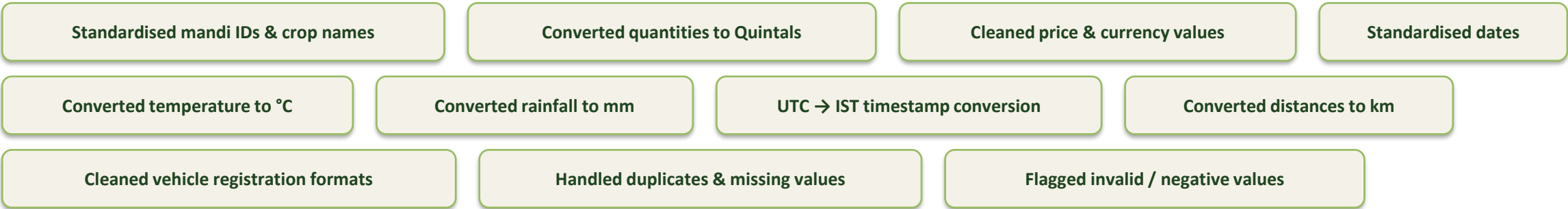
Transport Logistics

Transit time, distance and warehouse routing

From Raw to Reliable



KEY CLEANING OPERATIONS



How We Score Risk

Every mandi receives three sub-scores, combined into one weighted Priority Score.

50%

Price Risk

30%

Supply Pressure

20%

Transport Risk

= Priority Score → Critical · High · Medium · Low

Priority Score Weighting

- Price Risk: 50%
- Supply Pressure: 30%
- Transport Risk: 20%

What the Data Shows

25,000

mandi arrival records analysed

10,164

unique date-mandi-crop price records

3,079

price observations recorded below MSP

42.05%

wheat price-crash rate — highest of all crops

9,862

valid transport records analysed

26.57%

transport trips delayed beyond 18 hours

Average modal prices stayed above MSP for all six crops overall — but individual observations frequently fell below it. Daily rainfall and arrivals showed almost no correlation (0.022).

Mandis Needing Attention

Ranked by the weighted Priority Score — the five highest-risk mandis in the dataset.

RANK	MANDI	PRIORITY SCORE	STATUS
1	Orai Market	90.38	Critical
2	Khandwa Grain Market	89.09	Critical
3	Dehri Mandi	89.00	Critical
4	Gurgaon Grain Market	87.93	Critical
5	Guntakal Mandi	87.65	Critical

Ask It Anything

A natural-language interface over the entire supply-chain dataset — no code or database queries required.

1

Trend analysis of arrivals & prices

2

Crop, mandi & district comparisons

3

Price and MSP analysis

4

Supply analysis from arrival quantities

5

Weather analysis by location & time

6

Transport analysis — transit & delays

7

Warehouse-level crop movement

8

Risk & priority analysis

Depending on the query, the agent returns numbers, comparisons, tables or visualizations — a natural-language exploration layer over the whole analysis.

Built With

LIVE DASHBOARD

datathon-agritech.vercel.app

Data Analysis

- Python
- Pandas
- Jupyter Notebook

Data Sources

- CSV
- JSON

AI Agent & App

- Python
- JavaScript
- HTML

Dashboard

- Web Dashboard
- Vercel

Development

- VS Code
- Git
- GitHub

What's Next

Directions to extend the project beyond the datathon.

1

Real-time mandi prices

2

Live weather updates

3

GPS-based transport tracking

4

Crop price forecasting

5

Route optimisation

6

Automated price-crash & delay alerts

7

More advanced natural-language queries

8

Deeper district & crop-level recommendations

Thank You

Mandi-to-Market Supply Chain Optimizer — TransOrg AgentIQ Datathon, Track 3: AgriTech

Live dashboard: datathon-agritech.vercel.app

GitHub: github.com/vanshika-vermaa/datathon-agritech

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